

Multiple Choice (10 marks)

1. Scientists are concerned about the presence of heavy metals like lead and mercury in the environment because
 - (a) heavy metals threaten some forms of life.
 - (b) heavy metals will increase available resources.
 - (c) exposure to heavy metals causes healthy mutations.
 - (d) mining for heavy metals causes environmental stability.
2. Which of the following is least likely to allow electricity to flow?
 - (a) aluminum
 - (b) copper
 - (c) plastic
 - (d) water
3. What is a benefit of the regulation of gene expression?
 - (a) conservation of genetic information
 - (b) conservation of cell resources
 - (c) trait adaptation to environmental change
 - (d) trait inheritance in offspring
4. Which is most likely a learned behavior for a raccoon?
 - (a) making a den in the hollow of a tree
 - (b) opening a trashcan looking for food
 - (c) moving four legs while running
 - (d) having a litter of four babies
5. Coal is formed from
 - (a) seas that have evaporated.
 - (b) natural gas subjected to heat.
 - (c) melted and cooled metamorphic rock.
 - (d) plant remains decomposed under pressure.

True / False (10 marks)

Answer True or False

6. True or False: Dough bakes in an oven, resulting in a chemical change as it transforms into bread.
7. True or False: The equation $2\text{C}_6\text{H}_6 + 7\text{O}_2 \rightarrow 4\text{CO}_2 + 6\text{H}_2\text{O}$ demonstrates the law of conservation of mass during the combustion of ethane.
7. True or False: A solution of water and a nontoxic compound can be safely poured down a drain after the demonstration is complete.
8. True or False: A scientist needs accurate records that show the data are consistent to prove a discovery is valid.
9. True or False: Plants wilt on hot days primarily because they are actively engaging in photosynthesis at a high rate.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the roles of chloroplasts and vacuoles in plant cells, focusing on how they store energy from sunlight and how this energy is utilized by the cell.
12. Describe the methods the students could use to differentiate between table salt and table sugar, emphasizing the importance of accurate labeling and identification in scientific investigation.

End of Paper